

Integration capabilities: It is compatible with telephony systems, CRMs, and databases, enabling a unified communication platform.

Community support: A large open source community provides updates, security patches, and continuous development.

Scalability: Asterisk can scale with business growth, supporting everything from basic setups to large-scale call centres.

Ownership and control: On-premises or private cloud setup ensures full data ownership, unlike vendor-controlled proprietary solutions.

Asterisk is indeed a great choice for setting up a call centre, especially for those familiar with basic Linux and networking. Its flexibility, active community, and open source nature make it accessible and adaptable, allowing for a tailored call centre setup without extensive technical expertise.

Getting started with Asterisk

To get started, you'll need:

- **A server:** Preferably with a multi-core processor and at least 4GB of RAM for optimal performance.
- **A Linux-based OS:** CentOS or Debian are popular choices, but any distribution compatible with Asterisk will work.
- **Basic Linux and networking knowledge:** Understanding of command-line operations will help with installation and troubleshooting.

We can install and run Asterisk on Ubuntu 24.04. It can be installed either by using the source code or through the package manager.

Installing Asterisk from source offers several advantages over using package managers like apt-get, particularly if you're looking for flexibility, customisation, and up-to-date features. Here's why installing Asterisk from source is often recommended.

Customisation and configuration: When installing from source, you have complete control over which modules and features to include or exclude. This is particularly useful if you only need specific components of Asterisk for a specialised setup, allowing for a leaner and more optimised installation.

Latest features and security updates: The apt-get repository often has older versions of software. Compiling from source allows you to install the latest version of Asterisk, ensuring you have the latest features, performance improvements, and security patches.

Compatibility with other software: Some dependencies or modules may need specific versions that are not available in the default package repositories. Installing from source allows you to manage these dependencies more flexibly.

Optimisation for specific hardware: When you compile

from source, you can optimise the build for the specific hardware or operating environment. This can improve performance, especially on resource-limited systems or high-traffic call centres.

Control over dependencies: Asterisk has many dependencies, and package managers like apt-get may install versions that are outdated or incompatible with your other software. Compiling from source lets you control exactly which versions of libraries (like libpri or dahdi) you use.

Better debugging and development options: If you're actively developing or debugging, a source installation offers better options for compiling with debugging symbols. This allows you to track issues more effectively or integrate custom patches for experimental features.

However, installing from source requires a bit more manual effort and an understanding of Asterisk's dependencies and configuration options. If you need a stable, less customisable setup and are fine with potentially older versions, apt-get is a more straightforward option.

Now follow these steps:

1. Update Ubuntu
 2. Install pre-requisite library
 3. Download asterisk source
 4. Configure
 5. Configure and install asterisk
- Each step is elaborated as under:

```
apt-get update
sudo apt install -y build-essential libncurses5-dev libssl-
dev libxml2-dev libsqlite3-dev uuid-dev libedit-dev mpg123
libcurl4-openssl-dev
cd /usr/src
sudo wget https://downloads.asterisk.org/pub/telephony/
asterisk/asterisk-20-current.tar.gz
sudo tar -xvf asterisk-20-current.tar.gz
cd /usr/src/asterisk-20*
./configure --with-jansson-bundled (See fig conf_complete.png)
make menuselect #select or deselect any module in asterisk
make # to compile source code (see fig compile_success.png)
make install
make samples
systemctl start asterisk
asterisk -rvvvvv (see fig asterisk_console.png)
```

The setup is complete and we will now configure it to make a test call.

There are two files you need to learn and practice -- *pjsip.conf* and *extensions.conf* — which are located in the */etc/asterisk* directory.

We will first create two extensions 601 and 602. We will open */etc/asterisk/pjsip.conf* and add the code given below at the bottom of this file: